दक्षिण बिहार केंद्रीय विश्वविद्यालय, गया

**CENTRAL UNIVERSITY OF SOUTH BIHAR**

NH-120, Gaya- Panchanpur Road, Village- Karhara, Post- Fatehpur
P.S- Tekari, District- Gaya (Bihar) PIN- 824236

## Question Paper of End –Term Examination May 2026

**Name of Department**: Deptt. of Computer Science **Programme**: M.Sc. (AI)

**Session**: 2025-2027 **Semester**: 2nd

**Course Title: Wireless Sensor Networks** **Course Code**: CAI82DE01304

**Course Credits**: 4 **Max. Marks**: 70 **Duration**: 3 Hrs

**Enrolment No. (to be filled by Student)**:…CUSB2SD2332012……………………..

### Instruction:

1. Attempt questions from all the Sections as directed.

### Section - A

***Answer all questions below, each carrying 1 point.*** **[10 X 1 = 10]**

1. Define a transducer.
2. What is WSAN?
3. List the names of two Optical Sensors.
4. List the types of coverage you can have in a WSN.
5. What is a Flat Wireless Sensor Network?
6. What is a routing hole?
7. What if the probability of transmission p is chosen near 1 in Gossip Routing?
8. Can a neighboring relation be asymmetric? Explain briefly.
9. What do you mean by overhearing?
10. What is eavesdropping?

### Section - B

***Answer any 5 questions below, each carrying 6 points.*** **[5 X 6 = 30]**

1. What are the main differences between MANET and WSN?
2. What are the active sensors and passive sensors? Discuss with an example.
3. Discuss the virtual force-driven deployment scheme of WSN.
4. What is localization? Why is it needed in WSNs? [2+ 4=6 Points]
5. What are the key challenges of neighbor discovery problem?
6. What are the major sources of energy wasted in Sensor nodes in the context of MAC protocols?

## Section - C

***Answer any 2 questions below, each carrying 15 Points.*** **[2 X 15 = 30]**

1. What do you understand by negotiation-based routing protocols? Discuss any two protocols from the SPIN protocol family. [2+ 13=15 Points]
2. Write a short note on the following: [7.5+ 7.5=15 Points]
   a. Active Sensor Model for Dynamic Sensor Network Programming
   b. S-MAC protocol
3. Answer the following:
   a. Discuss the CIA model of Security in the context of WSN. [7.5 Points]
   b. Discuss engineering and agricultural applications of WSNs in detail. [7.5 Points]

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